

Title: Transformer Models for Protein Folding

Abstract: We present FoldNet, an end-to-end transformer model for predicting 3D protein structure from amino a

Section 2: Problem & Methodology

Existing MSA-based methods suffer from high compute cost $O(N^3)$. FoldNet uses axial attention to achieve $O(N^2)$.

Section 4: Results & Future Work

FoldNet achieved 88.4 GDT score on CASP14 benchmark. Limitation: Requires high GPU memory. Future Work